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Food Beliefs and Practices During Pregnancy in Ghana: Implications for Maternal Health Interventions

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Food Beliefs and Practices During Pregnancy in Ghana: Implications for Maternal Health Interventions

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Ghanaian women's food beliefs and practices during pregnancy and the scope for developing more effective maternal health interventions were explored in this study. Thirty-five multiethnic Ghanaian women between the ages of 29 and 75 were interviewed about pregnancy food beliefs and practices. I show that, based on the data analysis, their knowledge about food was drawn from lifeworlds (family and friends), educational settings, health professionals, mass media, and body-self knowledge (unique pregnancy experiences). Core lay ideas converged with expert knowledge on maternal health nutrition. Multiple external factors (e.g., economics, cultural representations of motherhood) and internal factors (e.g., the unpredictable demands of the pregnant body) influenced pregnancy food practices. I suggest and discuss a need for culturally situated multilevel interventions.

Ghanaian women of childbearing age experience problems with nutrition. Poor women experience undernutrition; wealthier women experience overnutrition. These problems have implications for maternal and child health. Maternal nutrition interventions in Ghana are informed by a common assumption that lay nutrition knowledge is "incorrect" and expert nutrition knowledge is "correct." By pitting "good" expert knowledge against "faulty" lay knowledge, researchers and health practitioners fail to incorporate the complex range of beliefs, knowledge, and practices that shapes women's pregnancy experiences and food practices. In this study I apply the concepts

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of medical logic and social logic: (a) to examine the content, sources, and functions of women's beliefs about pregnancy foods; (b) to identify food practices during pregnancy and their mediating factors; and (c) to examine the implications for developing more appropriate dietary interventions for pregnant women. Interviews with 35 women of different ethnicities and socioeconomic status suggested that successful maternal interventions will have to consider the following: (a) pregnant women draw on multiple and often-conflicting sources of information; (b) lay and expert knowledge converge on core ideas of maternal health and antenatal nutrition; and (c) the pregnant body is a significant mediator of pregnancy food practices.

Ghanaian women of childbearing age from poor rural and urban communities experience food and micronutrient insecurities (Grant & Lartey, 2006; Ghana Statistical Service [GSS], Noguchi Memorial Institute for Medical Research [NMIMR], & ORC Macro, 2004, 2009). These insecurities increase their risk of disease and premature death. In Ghana iron-deficiency anemia is a major threat to maternal health, contributing to low birth weight and poor health and cognitive outcomes for babies, as well as increasing morbidity risks for mothers. At least 9% of Ghanaian women are chronically malnourished: high mortality rates are reported among women from the northern belt of Ghana and of low education. The prevalence of anemia among Ghanaian women is higher than the prevalence in other West African countries, including Benin, Guinea, Sierra Leone, and Senegal (GSS et al., 2009).

Women with higher income and education in urban settings experience overnutrition. This group of women has access to a wider diversity of local and foreign foods, report increased intake of fat-rich and sugar-rich diets, and are more likely to be overweight or obese (Biritwum, Gyapong, & Mensah, 2005; Dake, Tawiah, & Badasu, 2010; GSS et al., 2009; Levine et al., 1999). Thirty percent of Ghanaian women are reported to be overweight or obese: the Greater Accra Region—where this study was conducted—has the highest percentage (45%) of this group. Overweight and obese women are more likely to experience pregnancy-induced diabetes and hypertension, which undermine their physical and psychological well-being during pregnancy and increase their chance of delivering overweight babies, who—current knowledge suggests—may go on to develop obesity and diabetes as adults. Pregnancy-induced hypertension is a leading cause of maternal death in Ghana (Kumi-Aboagye, 2008; Lassey & Wilson, 1998).

There have been two broad responses to the dual challenge of undernutrition and overnutrition among Ghanaian women of childbearing age. At the national level, nutritionists and food researchers have developed micronutrient supplementation and local food fortification projects as well as public health education strategies on malnutrition and overnutrition (Owusu, Adom, Opoku, Afoakwah, & Ankrah, 2006; Quarshie, Amoaful, & Armah, 2006). At the community level, micronutrient supplementation has become central to maternal health and antenatal care. At (public) antenatal clinics across the

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country pregnant women receive generic micronutrient supplements such as folic acid, multivitamins, and b-complex vitamins free of charge. For women who face pregnancy risks such as preeclampsia, hypertension, and diabetes, diet restrictions are advocated alongside micronutrient supplementation. Restrictions focus largely on reducing salt, fat, and sugar intake and necessitate restructuring everyday food practices. It is important to note that maternal health care in Ghana is generally poor especially at district and community health facilities, and many health facilities lack nutritionists and dieticians (GSS et al., 2009; Initiative for Maternal Mortality Programme Assessment [IMMPACT], 2007). Reports suggest that pregnant women using district and community health facilities may not necessarily have access to expert knowledge on maternal nutrition. Current interventions have had mixed success. At the community level, women have not fully adopted the practice of taking daily micronutrient supplements although they understand the benefits of these supplements to their reproductive health. Similarly, compliance with antenatal micronutrient supplement intake (such as vitamin A and iron) and dietary advice across the country is reportedly low (GSS et al., 2009). Local experts advocate for the development of more effective nutrition interventions (Addo, Marquis, & Lartey, 2006; Quarshie et al., 2006).

In this article I report a qualitative study that examines pregnancy food beliefs and practices in Ghana. This study is a subset of a broader study reported elsewhere that focused on the role of culture in pregnancy and breastfeeding dietary practices (de-Graft Aikins, 2011). The study is based on the premise that nutrition interventions may not be successful because they fail to incorporate women's perspectives on food and nutrition and their daily food and nutrition practices. This approach is informed by a tendency to pit "good" expert knowledge against "faulty" lay knowledge. I had four aims: (a) to document the content, sources, and functions of women's beliefs about pregnancy foods; (b) to identify food practices during pregnancy and their mediating factors; (c) to examine the extent to which these representations and practices corresponded to or deviated from expert knowledge; and (d) to identify practical and effective dietary interventions for pregnant Ghanaian women.

CONCEPTUAL FRAMEWORK

Maternal nutrition interventions in Africa are informed by a common assumption that lay nutrition knowledge is "incorrect" and expert nutrition knowledge is "correct" (Appoh & Krekling, 2004; Gittelson & Vastine, 2003). Writing within the Ghanaian context—on maternal nutrition knowledge in the Volta Region—Lily Appoh and Sturla Krekling, state that "whereas nutritional knowledge obtained from formal education and community health services to a large extent may be relied on to be the right one,

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the same cannot be said of knowledge about nutrition obtained through friends and families that may be related to the culture, tradition and beliefs in the community" (2004, p. 1). The popular, but conceptually limiting, knowledge–attitude–behavior (KAB) model of health promotion posits that knowledge changes attitudes, which in turn change behavior (Joffe, 1996). Maternal health experts often draw explicitly or implicitly on the KAB model and assume that providing "good" expert nutrition knowledge to women who desire healthy pregnancies and babies enhances their ability to achieve healthy status and therefore should ensure adherence with nutrition interventions. Yet interventions have had mixed success.

The conceptual framework for this study was informed by the concepts of "social logic" and "medical logic." "Social logic" and "medical logic" were coined within the health psychology field to distinguish between the knowledge drawn on by people with illness and their caregivers to manage their condition and the knowledge drawn on by their health care providers (Herzlich & Pierret, 1987; Nettleton, 1995). "Medical logic" is drawn from the disease-centred approach to illness and focuses on a restricted repertoire of practical routines aimed at addressing the physiological dimension of the illness. "Social logic" draws from intersubjective experiences of the individual with illness and focuses on a broader repertoire of practical routines aimed at addressing the physiological as well as social dimensions of living with illness. Social logic will include what Appoh and Krekling refer to as "knowledge ... obtained through friends and families which may be related to the culture, tradition and beliefs in the community" (2004, p. 1). Researchers argue that the everyday knowledge and skill drawn from "social logic" confer expertise to the chronically ill and their caregivers that encapsulates and transcends the expertise of their health care providers; this expertise has to be brought to the fore (and legitimized) in trying to understand processes of medical and self-care. These observations are true for experiences of chronic diseases such as diabetes and cancer in Ghana (Clegg-Lamptey & Hodasi, 2007; de-Graft Aikns, 2003; de-Graft Aikins et al., 2014).

When applying this broad argument to maternal health, two corresponding scenarios can be developed. Expert knowledge on maternal nutrition, hereafter "medical nutrition logic," can be described as highly technical knowledge based on nutritional science and obstetrics that focuses on a restricted repertoire of practical nutrition management routines aimed at addressing the physical and physiological dimensions of pregnancy. On the other hand, lay knowledge on maternal nutrition, hereafter "social nutrition logic," draws on physiological, physical, as well as intersubjective and sociocultural dimensions of pregnancy experiences and as such instructs a broader repertoire of practical food and nutrition practices. Social nutrition logic essentially captures what pregnant women know, think, feel, and do about food during their pregnancies. This goes beyond the medical space and encompasses the social, cultural, and structural dimensions of everyday life.

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TABLE 1 Participant Profiles

Participant information	No. of participants
Age	
21–40	13
41–60	14
61–80	8
>80	
Ethnicity	
Akan	13
Dagao	3
Ewe	2
Ga	5
Hausa	3
Education	
None	1
Primary	2
Secondary	7
Tertiary (polytechnic, vocational training) University	5
Occupation	
Professional (law, management, academic)	5
Professional (nursing, teaching, secretarial, support services)	5
Trader (market, store)	6
Retired (but working—consulting, running kindergarten, etc.)	3
Retired (no work)	1
Other (e.g., traditional leader)	6

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METHODS

Thirty-five women of different ethnicities (Akan, Dagao, Ewe, Ga, and Hausa) and age groups (29 to 75) were recruited for the study. The study was exploratory, and the broader focus was on the role of culture in pregnancy and breastfeeding experiences. Therefore, I aimed to maximize diversity across ethnic identity and age in order to explore cultural differences (via ethnic difference) as well as the changes in the role of culture over time (via intergenerational experience). In Table 1 I present the profile of the study women. With the exception of one, all informants were educated. Educational status ranged from primary-level education to obtaining a postgraduate university degree. Occupations ranged from small-scale trading to white-collar professions. Some retired women continued to work. The number of children women had ranged between one (for the youngest informants) and four. The youngest age at first birth was 20 (for a woman in her late forties); the oldest age for first birth was 36 (a woman lactating at the time of the study).

After gathering demographic information (age, education, occupation[s], number and ages of children, ethnicity), the women were asked questions about the following:

(1) recommended and prohibited pregnancy foods, including sources of knowledge and functions of the foods;

(2) foods consumed during their own pregnancies and the factors mediating food practices; and
(3) their general health and well-being during pregnancy.

All the women generated their pregnancy food list using a free-recall approach. Question (1) produced a general pregnancy food list, and question (2) generated a participant-specific pregnancy food list. As interviews progressed and perspectives approached "meaning saturation" (the "stopping criterion" for qualitative research: the point at which no additional variety of meaning is gained from data collection (Gaskell, 2000), items on the cumulative pregnancy food list were used as memory prompts (for example, "I spoke to some women who said snails were good for pregnancy, what is your opinion on that?"). A key limitation of this open-ended approach with a cohort of women who were required to remember their pregnancy experiences dating back a number of years or decades was that certain details of everyday pregnancy experience such as food portions, quantity of herbs used in cooking, and frequency of food intake were lost. This kind of detailed information is best gathered by direct observation or the use of food diaries and would necessitate studying pregnant and lactating women over extended periods of pregnancy and breastfeeding (Messer, 1989). I interviewed the women along with two research assistants in English and the local languages: Asante-Twi, Fante-Twi, Ga, and Dagari. The interviews were conducted between October 2006 and April 2008. Notes were taken for all interviews; some women were interviewed more than once. Note-taking was preferred because it created a more relaxed interviewing setting, especially for opportunistic interviews. For some interviews, note-taking was the only practical option, for example, with market women interviewed during work. Using the operationalised notion of "social nutrition logic" analysis focused on content, sources, functions, and legitimation of food knowledge.

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RESULTS

Results are presented in two parts. In Part 1 I focus on women's knowledge of recommended and prohibited foods for pregnancy. In Part 2 I focus on food practices during women's pregnancies. Cross-cutting themes include women's relationship with food during their pregnancies and the extent to which social nutrition logic structures everyday food practices and enhances or undermines compliance to expert nutrition interventions.

Part 1: Social Nutrition Logic: Contents, Sources, Functions, and Legitimation of Pregnancy Food Beliefs

Contents of social nutrition logic. The women generated a wide range of recommended foods and food supplements for pregnancy. The majority

of foods were indigenous to Ghanaian diets generally (e.g., green leafy vegetables, legumes, meat and fish, fruits). These indigenous foods are termed traditional foods and supplements: their indigenous status allows identification of continuities and change in culturally mediated food consumption patterns. The list also included foreign foods such as beverages (tea, Milo, Ovaltine, Horlicks), dairy products (milk, yogurt), and biscuits (digestives, cream crackers). The contents of the food list, including incorporation of foreign foods, was similar across different age groups. That is to say pregnancy diets had not changed much across three generations (between women in their late sixties, forties, and twenties). The majority of indigenous foods discussed cut across ethnic groups. Two notable, but expected, differences occurred in terms of relative food availability (e.g., more fish in southern Ghana, more meat in the northern Ghana) and mode of food preparation (e.g., preference for fermented maize/millet dough-based staples in some ethnic groups and nonfermented maize/millet dough-based staples in

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TABLE 2 Recommended Food and Food Supplements for Pregnancy

Raw foods	Cooked foods	Dry/processed foods
Fruits: bananas, oranges, pawpaw (papaya), pineapples, watermelon	Beans stew (with palm oil) [eaten with boiled white rice]	Beverages: tea, Milo, cocoa
Vegetables: carrots, cabbage, lettuce, "salads"	Bread (white, brown)	Biscuits: cream crackers, digestive
Honey	Fish: fried *keta school boys* (anchovies), smoked mackerel [with fresh pepper sauce, kenkey[ii]]	Fluids: soya milk, juices
Supplements: *kwawu nsua/nsaman troba* [i] (blended with water)	*Nkontomire* stew (made with palm oil) [with boiled sweet cassava[iii], or plantain, or yam]	Supplements: blood tonics; b-complex vitamins, biscuits (dry) folic acid, lucozade, multivitamins, pregnacare
	Palm-nut soup [with fufu[iv], banku[ii], rice]	
	Soups (other)—light (tomato & garden eggs); nkontonmire vegetable[v]	
	Snails (in soups)	
	Stews (other): garden eggs, Tuo Zaafi[v] [with vegetable soup]	
	Supplements: *alefi*[v]; *kwawu nsua/nsaman troba* (added to soups and stews); prekese	

Ethnic differences: [i]mentioned only by Akan women; [ii]made from fermented corn dough—common to southern women (Akan, Ga); [iii]common to Nzema and Krobo communities; [iii]usually pounded plantain and cassava, mentioned only by southern women; [v]mentioned only by northern women.

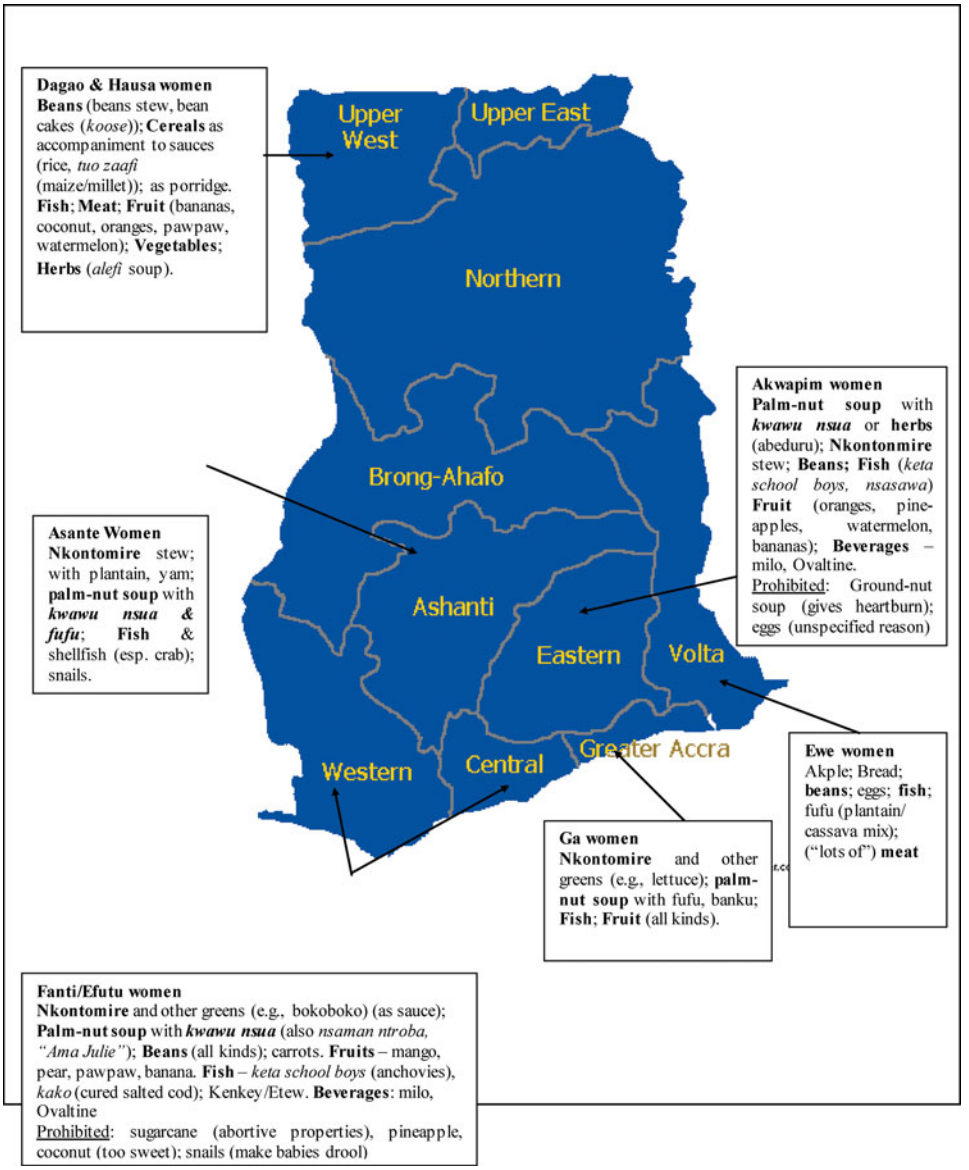


FIGURE 1 Recommended and prohibited pregnancy foods by ethnic group.

others). In Table 2 I present the cumulative pregnancy food list generated by the women under three categories: (a) raw foods, (b) cooked foods, and (c) dry/processed foods. Foods tied to specific ethnic groups are highlighted. In Figure 1, where I present a map of foods generated by ethnic group.

Some women made reference to prohibited foods. There was general agreement that excess fat and sugar was bad for pregnancy. Beyond this, reference was made to specific foods by Fante and Akwapim women. Some

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TABLE 3 Prohibited Foods and Food Supplements for Pregnancy

Raw foods	Cooked foods	Dry/processed foods	Nonfoods
Fruits: oranges, pineapple, coconut, sugarcane	Ground-nut soup	High-fat; high-sugar foods	Clay (Akwapim)
	Eggs Chilli pepper (in cooked sauces)	Cooking oils	
Chilli pepper (in fresh pepper sauces)		Salt	

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Akwapim women stressed that groundnut soup caused heartburn and was therefore prohibited in pregnancy. The status of other prohibited foods was not as clear-cut. For some women eggs were prohibited in pregnancy: for others eggs were recommended. One Fante woman placed extremely sweet fruits—sugarcane, pineapple, coconut—under the prohibited foods list. She observed that sugarcane had abortive functions and had to be restricted during pregnancy: her view was not common. A final category of prohibited "foods" included items such as clay and ice: their consumption by pregnant women in a variety of settings has been attributed to pica: "a perverted appetite for substances not fit as food or of no nutritional value" (Goldstein, 1998, p. 465). Because these items belong to the participant-specific pregnancy food list (i.e., food that is craved, consumed, or both during pregnancy but may not be generally recommended), it is more appropriate to discuss the context of consumption in Part 2. In Table 3 I present the list of prohibited foods.

Sources of Social Nutrition Logic

Women drew their knowledge of pregnancy and breastfeeding foods from five sources:

(a) family and friends, hereafter "lifeworlds";
(b) educational settings, typically school and university;
(c) health systems: the biomedical sphere (doctors, nurses) and traditional herbalists;
(d) media (newspapers, radio, and television); and
(e) unique pregnancy experiences (hereafter, "body-self knowledge").

The dimensions of knowledge from lifeworlds and pregnancy experiences are worth elaborating on, as these constitute the core of "social nutrition logic" and have been labelled "faulty" by nutrition experts. Knowledge drawn from lifeworlds can be categorized as cultural knowledge. The

knowledge was handed down from generations through mothers and other female members of the extended family as well as from friends and close community members. This knowledge encompassed physiological, physical, as well as intersubjective and sociocultural dimensions of pregnancy experiences and as such instructed a broad repertoire of food and health practices. It was ethnically bound, so that foods specific to certain groups and regions of Ghana appeared on the list of women originally from these regions, however long they had lived in Accra (see Figure 1). All women cited their lifeworld as a primary source of knowledge for managing pregnancy.

Body-self knowledge has been defined as subjective knowledge of one's unique state of physical and psychological balance (Helman, 2000). Body-self knowledge has been reported to mediate health maintenance and illness management practices in a variety of contexts including in Ghana (Angel & Guarnaccia, 1989; Bates, Rankin-Hill, & Sanchez-Ayendez, 1997; de-Graft Aikins, 2003; de-Graft Aikins, Awuah, Pera, Mendez, & Ogedegbe, 2014). Body-self knowledge emerged as an important source of knowledge during pregnancy. This knowledge modality was particularly important for women with more than one child because body-self knowledge shaped food practices throughout all pregnancies and to a large extent mediated use of other knowledge modalities.

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Functions of Recommended Foods

Most women attributed two dominant nutritional functions to the listed foods. Some foods, such as green leafy vegetables (consumed as stews or soups), were understood to "give blood" (37-year-old Asante woman) or prevent anemia. Others, such as palm nut (consumed as soups), were understood to "give body" (37-year-old Asante woman) or strengthen the pregnant body. Herbs and other food supplements such as *prekese (Tetrapleura tetraptera*), *alefi*, and *kwawu nsua* were central to meals and were believed to augment the processes of giving blood and body or minimizing physiological disruptions such as nausea (see Part 2).

In Table 4 I juxtapose the functions attributed to traditional foods with functions attributed to these foods by local nutrition experts. Three nutrition experts were consulted in producing social nutrition logic–medical nutrition logic comparisons for Table 4.

Functions attributed to pregnancy foods by the women were similar to functions attributed by nutrition experts. The traditional foods and supplements listed functioned to prevent anemia, ensure physical strength, minimize physiological disruption, and ultimately maximize the health of babies. The functions of some foods—fruits, vegetables—were not explicitly described. Their inclusion within the recommended pregnancy food list by the majority of women, however, suggests that they perform taken-for-granted maternal health functions. These foods are themselves recommended by

TABLE 4 Functions of Pregnancy Foods and Food Supplements: Comparing Social Nutrition Logic and Medical Nutrition Logic

Social nutrition logic	Foods and food supplements	Medical nutrition logic
Foods that prevent anemia ("giving blood")	Leafy green vegetables (e.g., nkontomire); other vegetables (*kwawu nsua*)	Sources of Vitamin A & C (for healthy bones) and folic acid (formation of new cells, production of red blood cells, protein synthesis)
Foods that strengthen the body & increase fertility ("giving body")	Palm-nut soup; fish, shellfish (esp. crab) and meats; snails; blood tonics; clay; beverages (tea, milo, Ovaltine with milk & sugar)	Sources of protein (vital to formation and growth of fetal brain), iron (used by body to increase maternal hemoglobin) & calcium.
Foods that minimize physiological disruption (nausea, vomiting)	Kola nuts	High source of caffeine; prohibited in pregnancy.
Foods that maximise health of fetus	Honey	Source of fructose, a carbohydrate: fructose has lower energy value compared with sucrose, but no direct link to fetal health
Foods with uncategorized functions	Vegetables (carrots); fruits (oranges, watermelons, bananas)	Carrots source of beta-carotene (which transforms to Vitamin A); Oranges & other citrus fruits source of Vitamin C.
	Accompaniments to sauces: fufu, kenkey, banku (or the Efutu *Etew*) TuoZaafi	Fermented maize/millet meals (e.g., kenkey) increase bioavailability of B vitamins, which regulate nerves (B1), maintain mucous membranes (B2), facilitate formation of red blood cells (B6, B12)

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nutritionists and maternal health experts as good sources of Vitamins A and C. Traditional sauce accompaniments such as *fufu*, *kenkey,* and *Tuo Zaafi,* whose functions were not outlined by the women, also have some nutritional functions as described in Table 4.

Part 2: Food Consumption Practices During Pregnancy: Physiological Changes, Bodily Demands, and Maternal Nutrition Compliance

Most women experienced physiological changes during pregnancy. Memories of these changes were acute, were linked to clearly demarcated periods

in pregnancy (e.g., "first 3 months," "first 5 months," "final month") or gender of unborn child (it felt different carrying boys vs girls for some), and constituted the lens through which relationships with food was discussed. Pregnancy was a period when many gained intimate knowledge of their bodies and physiological processes. For those who had more than one child, the first pregnancy provided a sensory map for future pregnancies. Some sensed they were pregnant before conclusive medical tests because their bodies underwent clear predictable changes; others knew they were expecting girls or boys.

For many women these physiological changes were primary mediators of food consumption during pregnancy. That is to say, depending on the severity of physiological changes, one could adhere to a well-balanced nutritious diet or not. Physiological changes mediated food consumption in three main ways, corresponding to three common experiences described by the women.

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Hypersensitivity and Hyperemesis

About half of the women had experienced hypersensitivity to smell, light, and motion during the first trimester. Hypersensitivity to smell was linked to nausea, spitting, and vomiting. Women who experienced nausea in their first trimester were therefore unable to eat a balanced diet because the smell of cooked (especially fried), spicy, or both foods exacerbated nausea. Most listed a restricted range of simple—"clean" (raw, steamed, boiled, roasted)—foods they could consume during this period (see Table 5). During this period, food supplements provided by antenatal clinics become central to maternal health.

TABLE 5 Foods Consumed During the First Trimester

Raw foods	Cooked foods	Dry/processed foods
"Fruits — oranges, pineapples, watermelon, and food supplements" (44, Akwapim)	"*Kobi**, *kpakpo shito***, okro (steamed)...kola nuts" (54, Ga) "pepper, fish (either fried or smoked), hot banku. Light soup. Tea with milk." (69, Ga) "fresh pepper, steamed crab, and kenkey; light soup" (47, Fante) "only dried fish" (75, Asante) "rice and bread" (69, Fante)	"Plain tea (without milk or sugar)" (64, Ga) "cereals... cocoa" (65, Krobo)
		Ice cubes and food supplements (37, Asante; 47, Akwapim) Soya milk, digestive biscuits, and food supplements" (36, Ga)

*Cured and salted tilapia; **small green aromatic chilli peppers.

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Increased Appetite and Craving Out-of-the-Ordinary Foods

Some women experienced increased appetite throughout their pregnancy or after the difficult trimester. During periods of increased appetite, most women ate all the recommended foods, often several times a day. For example, one first-time lactating mother (36-year-old Ga woman) recalled experiencing increased appetite after the first 3 months of (nausea-induced) limited eating. She ate heavy starch and protein-based foods on average seven times a day and at unstructured times and put on 3 kg of weight: her "doctor advised weight check." Weight gain was common during periods of increased appetite and posed health risks to mother and unborn child, according to the women's doctors.

Part of the experience of increased appetite was sudden and often prolonged craving for "out-of-the-ordinary" foods. A minority of women (across the age spectrum) craved and consumed pica: Akwapim clay and ice were two common substances. In Table 6 I present the list of craved foods.

The majority of foods belonged to the general list of recommended pregnancy foods. Of the remainder, one category—the "nonfood" items clay and ice—were not recommended within lifeworlds or by health care providers. As one elderly Fante women noted, "I used to eat clay—Akwapim clay. My body craved it. Nobody told us to eat it, we just wanted it."

The second category constituted foods such as ice cream and hamburgers and energy drinks such as "Malta Guinness" that, by virtue of their high-fat and high-sugar content, belonged to the prohibited food list. Craved foods served different purposes for different women: nutritional (e.g., calcium boost in Akwapim clay for one elderly Fante woman), physiological (the nausea-reducing properties of *bisi* [kola nut] for one middle-aged Ga

TABLE 6 Foods Craved by Women During Pregnancy

Raw foods	Cooked foods	Dry/processed foods	Other
Mangoes	*Akrantie* (smoked game)	Hamburgers	Clay
Kola nuts	Fish/shellfish: crab (steamed), *kobi*, *keta school boys*, *nsasawa*, shrimp (with light soup)	Ice cream	Ice cubes
		Malt drink	
		Yogurt	
	Game: akrantie (smoked)		
	Hausa *koko* (fermented maize/millet dough porridge with spices)		
	Hausa rice		
	Okra (steamed)		
	Rice, beans with vegetables		
	Snails (steamed, added to soups)		
	Yogurt		

woman) and psychological (the comfort provided by the smell, texture, and taste of *akrantie* [smoked game] for the elderly Fante woman).

Pregnancy-Induced Illness

Three women (ages 37, 45, and 46) developed hypertension during their pregnancies, three experienced persistent heartburn when they ate spicy foods, one woman (aged 44) developed preeclampsia during her second pregnancy, and an elderly Ga woman spent all of her four pregnancies in the hospital on a vitamin-enhanced drip. Most women in this subgroup were placed on strict—low salt, low fat, no spice—diet restrictions by their doctors during their pregnancies. These diet restrictions were often difficult to maintain for all the women: most struggled to lose weight. The remaining number of women reported having had "healthy pregnancies," including those who had hypersensitivity issues in their first trimester. Two women who had had pregnancy-induced hypertension had developed adult onset hypertension, which persisted at the time of interviews.

What becomes apparent from these accounts of pregnancy experiences is that bodily processes—and by extension the pregnant body—constituted an important legitimate source of maternal nutrition knowledge. As the elderly Ga woman who spent all her four pregnancies in the hospital on a drip observed, "We all know we need a balanced diet during pregnancy, but your body will dictate what you will eat."

DISCUSSION AND CONCLUSIONS

I had four aims, based on the premise that nutrition interventions in Ghana face challenges because they fail to incorporate women's perspectives on food and nutrition and their daily food and nutrition practices: (a) to document the content, sources, and functions of women's beliefs about pregnancy foods; (b) to identify food practices during pregnancy and their mediating factors; (c) to examine the extent to which these representations and practices corresponded to or deviated from expert knowledge; and (d) to identify effective dietary interventions for pregnant Ghanaian women.

According to medical nutrition logic, noncompliance is attributed to "faulty" cultural knowledge and poor access to expert nutrition knowledge and interventions. The focus on social nutrition logic, however, highlights the complex factors underpinning the relationships pregnant women have with food and by extension its consumption or nonconsumption. Three key insights emerge from this study:

1. Women draw on pluralistic knowledge sources to make sense of pregnancy and pregnancy diets, including lay and expert knowledge.

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2. Key elements of social nutrition logic and medical nutrition logic intersect. These include specific recommended and prohibited foods.
3. The pregnant body is a significant mediator of pregnancy food practices.

These findings have implications on developing effective antenatal nutrition interventions. In Figure 2, I present identified multilevel factors that mediate consumption of food and food supplements, including those advocated by maternal health experts. The current evidence suggests that medical nutrition logic is pitched solely at the structural/ideological level (lack of formal nutrition knowledge, lack of access to services) when it should also consider individual, interindividual, and group contexts of food practices. The study findings suggest that interventions that incorporate multilevel (structural, group, interindividual, individual) contexts of food beliefs and practices during pregnancy, but with sustained focus on the embodied experience of pregnancy, may yield better compliance outcomes. To conclude, I consider key challenges and opportunities for implementing multilevel interventions.

Pluralistic Knowledge on Pregnancy Foods and Nutrition

Most women drew on multiple sources of knowledge to make sense of pregnancy nutrition and food practices: lifeworlds, health systems, educational settings, the mass media, and, most importantly, their own pregnancy experiences. Processes of legitimation depended on whether the focus was on range and quality of knowledge and type and efficacy of nutrition intervention; these processes were rooted more broadly within changing

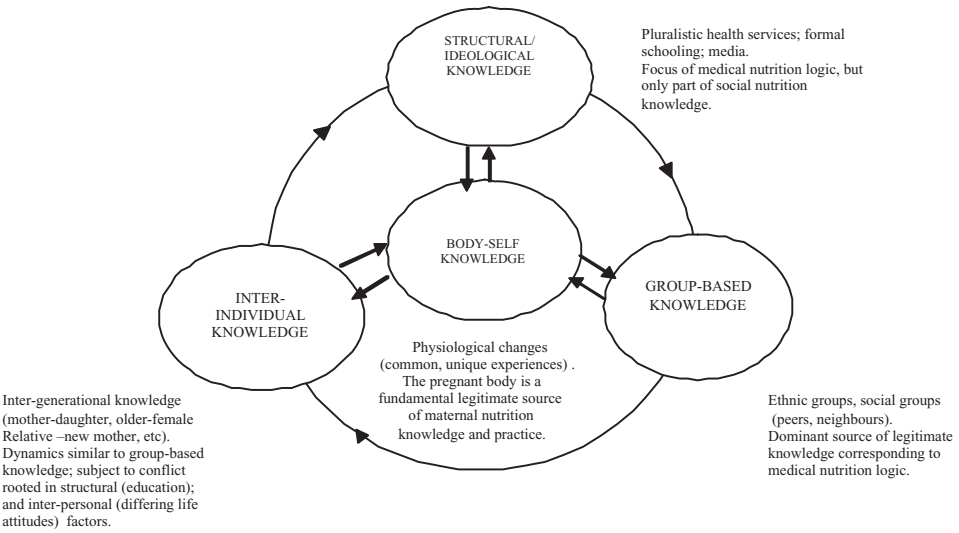


FIGURE 2 Multilevel factors mediating maternal nutrition practices.

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life—and unique pregnancy—experiences. Research on medical pluralism in sub-Saharan Africa suggests that individuals and communities perceive and use pluralistic health systems as coexisting alternatives (Ben-Tovim, 1985; Nyamwaya, 1987; Prins, 1992). Ghanaian research on medical pluralistic attitudes suggests that individuals and communities evaluate pluralistic medical systems along four criteria: technical or practical knowledge of health problems, technological expertise, accessibility (geographic, economic, cultural), and ethics (de-Graft Aikins, 2005). All systems have strengths and weaknesses across these criteria, depending on the health problem (de-Graft Aikins, 2005). Since medical pluralistic attitudes and practices dominate women's experiences, the way women of childbearing age position themselves toward pluralistic nutrition knowledge—rather than towards medical nutrition logic solely—constitutes a more critical area of future research.

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Social Nutrition Logic and Medical Nutrition Logic Overlap

The content of information from lifeworlds, which constituted the core of social nutrition logic (and deemed incorrect by nutritionists), corresponded in greater part to content of information from maternal health experts (deemed correct information). Five functional categories were given to recommended traditional foods and supplements. Of these, three categories corresponded to expert attributions: foods that prevented anemia (e.g., leafy green vegetables), strengthened the pregnant body (e.g., palm-nut soup and shellfish), and provided general health-enhancing functions (e.g., vegetables and fruit). Two categories were minority themes and did not have clear or positive functions according to experts: foods that minimized physiological disruption (kola nuts) and maximized the health of babies (honey). Identification of legitimation processes showed that women subjected all sources of pregnancy information to scrutiny. Thus elements of both social nutrition logic and medical nutrition logic that had proven functions were privileged.

Body-Self Knowledge: The Role of the Body in Pregnancy Food Practices

The most compelling common experience described by the women in this study was the role their bodies played in food consumption during their pregnancies. For many women the pregnant body constituted the ultimate guide to how one thought, felt, and acted about food daily during pregnancy. This guide over-rode other legitimate sources (e.g., structural, cultural) of knowledge on—the right or wrong—pregnancy foods. From the positive end, physiological changes in the pregnant body enhanced appetite for simple healthily cooked foods. From the negative end, extreme physiological changes caused either underconsumption or overconsumption of nutritious

or nonnutritious food. This common experience has two broad implications. First, it implies that embodied knowledge and its everyday functions have to be prioritized and legitimized in maternal nutrition research and in antenatal care in Ghana. Second, it suggests that pregnant women whose experiences lie at opposite ends of the physiological disruption continuum (e.g., loss of appetite vs. increased appetite) are more likely to be noncompliant with maternal nutrition intervention and therefore require greater individualized medical/nutrition support. The challenge lies in developing cost-effective and practical interventions that enable over-stretched and under-resourced maternal health services to provide such vital support.

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